THAPAR JAPAN HACKATHON 2026 · FINAL · PROBLEM JAL-PS-01

ThaliBento

थाली + **弁当** an Indian thali, packed with Japanese care

The airline keeps listening until the right meal reaches the right Indian passenger.

**“Not a new system.
It plugs into what JAL already uses.”**



A meal request is four letters on a booking. It falls off when the booking changes, and it can't hold how Indians actually eat. ThaliBento fixes both, and we'll show it live.

Team Caffeinated Nerds

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University) × JAPAN AIRLINES

GATE 02 THE PROBLEM

Today, the airline stops listening 25 hours before take-off

JAL asks about special meals once. But real life changes after that line, and that's where meals go wrong.

Before the line: one form, one 4-letter code. "No onion or garlic" or "only fasting on Purnima" has no code.

Special meals close at T-25 h

Allergy and kosher meals close even earlier: T-49 h

Booking
form + code

T-49 h
allergy line

Flight change
night before

Meal silently dropped. JAL's own page: reconfirm after any change.

Check-in / gate
airport

"It's my fast day." "I need a Jain meal." Too late.

Boarding

Paneer runs out at row 34. A meal is missing. No internet.

In flight

INDIA · DIET

81%

of Indians avoid some kind of meat, many only on certain days

INDIA → JAPAN · 2025

315,000

Indian visitors to Japan, 35% more than in 2024

NOBODY MEASURES

0

airlines that publish how often special meals fail

Sources: JAL special meals page (25 h / 49 h, reconfirm after changes), Pew Research Center (2021), JNTO via India Outbound

GATE 03 PROBLEMS AND FIXES

7 things that go wrong after the 25-hour cutoff

#	WHEN	THE PROBLEM	WITHOUT THALIBENTO	THALIBENTO
1	**17 hours** before take-off	A travelling companion asks for a Jain meal **after the cutoff**	No way to ask. The request goes nowhere	**Added to the kitchen's final list** (VJML), which is still open
2	**5½ hours** before take-off	A rebooking on a partner airline **silently deletes Priya's meal**	Nobody notices. No meal on board	**Caught and requested again** before the kitchen's final list
3	**3½ hours** before, at check-in	A passenger says she **can't have sugar** (medical)	"Too late to order a special meal"	**Diabetic meal from the kitchen's standby stock** before the trucks leave
4	**1¾ hours** before, at the gate	A passenger asks for a **Jain meal at the gate**	Met only if a spare happens to exist, unchecked	**A spare on board, checked against his rules,** held in galley 2
5	**45 minutes** before, at boarding	A passenger **doesn't show up**	Their meal is thrown away	**Their meal is freed** for the next person who needs it
6	**1½ hours** after take-off	A passenger **realises it's a fast day**	The veg meal has rice and lentils. Nothing fits	**Sealed universal plate** (samak millet, makhana, lauki), heated and served
7	**2 hours** after take-off	**Paneer runs out:** 3 meals left for 6 people	Last three passengers get nothing	**A flexible passenger takes chicken** (+300 miles), so 38G gets the paneer

Without: 1 of 5 strict needs met · 2 meals thrown away

With ThaliBento: 5 of 5 met · 0 thrown away · 104 diets known before the cutoff, not 24

Simulated JL740, Delhi → Tokyo; every row runs live in Chaos Mode

GATE 04 ONE PASSENGER

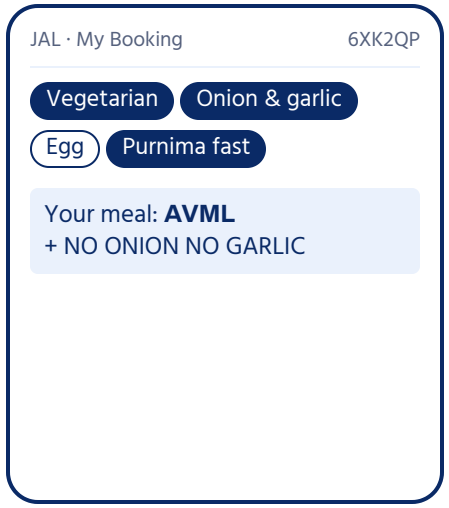
Priya's meal, from booking to seat 23C

Vegetarian, no onion or garlic, flying IndiGo into JAL's Delhi–Tokyo flight. Every screen here is in our live demo.

23 Oct 18:10 BOOKING

PASSENGER · PHONE

She ticks what she doesn't eat

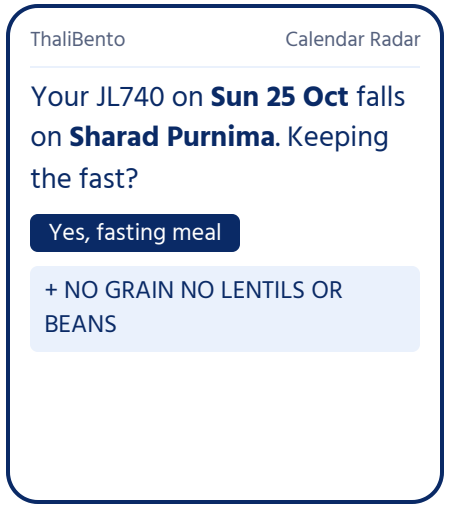


No codes to learn. We store food rules, never religion.

18:12 FAST DAY

PASSENGER · PHONE

A one-tap reminder

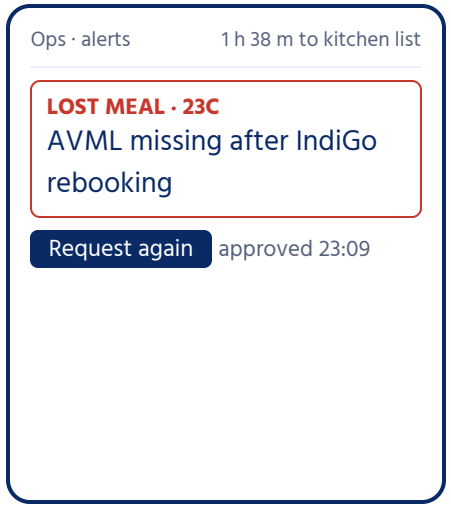


Caught a day early, not at the gate.

23:02 FLIGHT CHANGE

AIRPORT STAFF · LAPTOP

Her meal vanishes, we notice



Fixed in 7 minutes, before the kitchen closed.

00:40 KITCHEN

CATERER · LAPTOP

Clear label, hidden fish caught

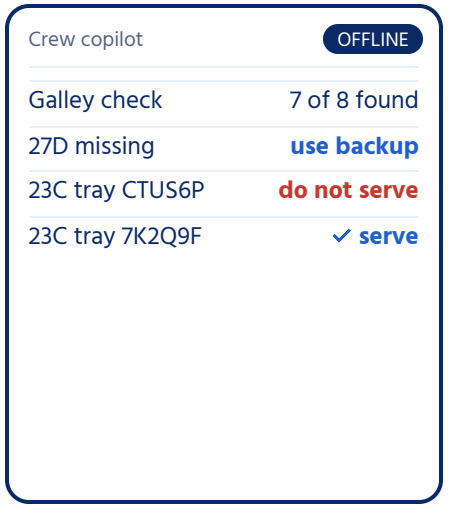


AI reads Japanese recipes; our rules decide.

25 Oct 06:30 ON BOARD

CREW · IPAD, OFFLINE

The right tray, checked

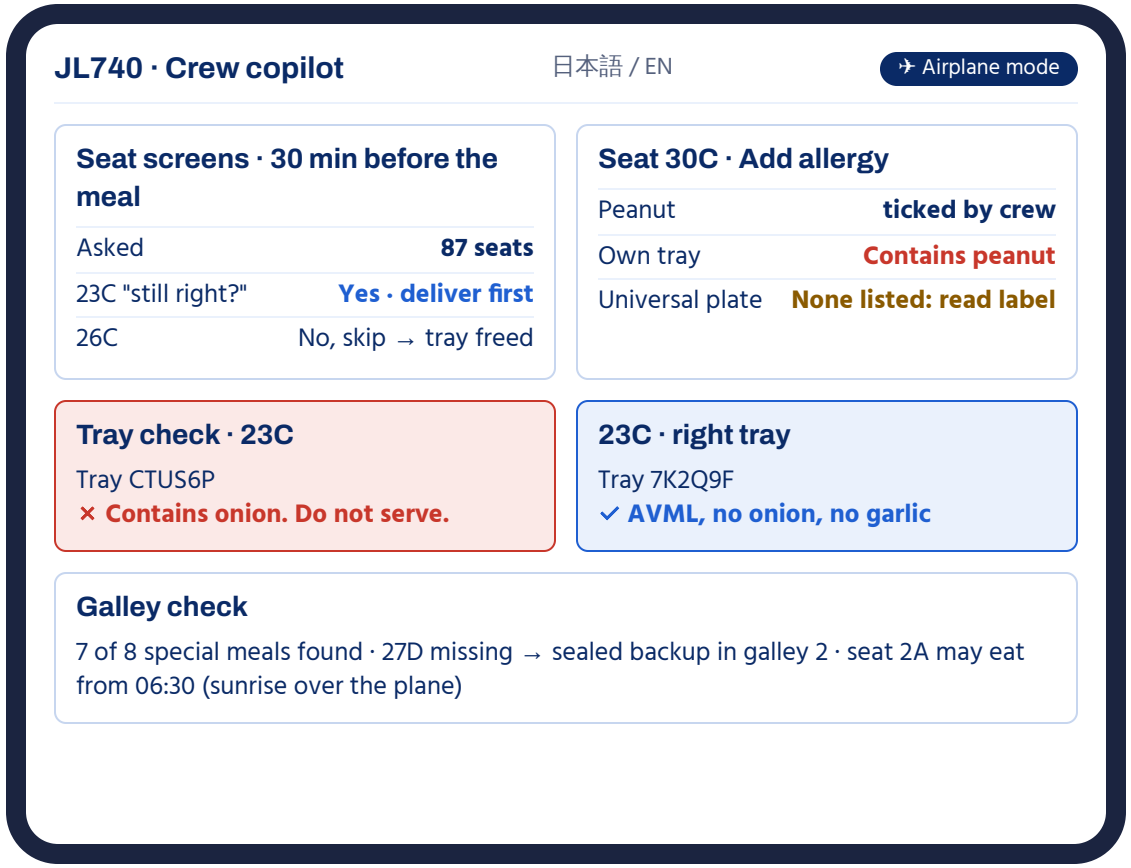


A wrong tray stops at the seat, no internet needed.

Priya is an example passenger on a simulated JL740; times are from our demo story

DEL ✈ NRT

In the cabin: four checks before anyone eats, all offline



1 **The seat screen asks first**
"Your pre-ordered meal: still right?" **Yes** goes on the crew iPad in 2 s. **No, skip** frees the tray for the next request. **Change** offers only dishes the rules allow.

2 **Allergy told on board**
Crew tick it on the seat card. Every dish on board is checked: **contains**, **none listed** or unknown. Never "safe".

3 **The tray check at the seat**
A tap doesn't prove who is sitting there, so the scan stays the final check: red or green, plus find by boarding pass after a seat swap.

✈ **No internet needed**
Everything downloads before take-off, works in airplane mode, and syncs after landing. Japanese and English.

Seat screens: JAL's 787s run Thales AVANT seatbacks with meal ordering; crew iPad: i-PAXINFO, offline

GATE 06 BUILT FOR INDIAN DIETS

What no meal code can say, ThaliBento handles

1 Diets that change by date

Priya's flight home is on Ekadashi: only that flight gets the fasting meal, with the note **in Japanese** for the Tokyo kitchen.

2 Eating by the sun

Sunrise over the plane comes **46 min** before Delhi's. Crew know when a passenger may eat; overnight flights get a sealed Sunrise Box.

3 Hidden ingredients in Japanese food

AI reads the caterer's recipe sheets (出汁 dashi = fish). **96%** alone, **100%** with our word list, on 23 sheets.

4 One meal for every diet

A sealed **universal plate** suits 13 of 13 diets on this route: **₹152 a flight**, vs ₹2,359 wasted with fresh spares.

5 Inside JAL's own booking

A short form in My Booking: tick what you don't eat. Allergies asked Yes or No first. The rules pick the code and kitchen note.

6 Lost meals caught

Every new passenger list is compared with the last. A meal dropped by a rebooking raises an alert, with the time left before the kitchen closes.

Same rules everywhere: 625 rule tests give identical answers on the server and the offline crew iPad.

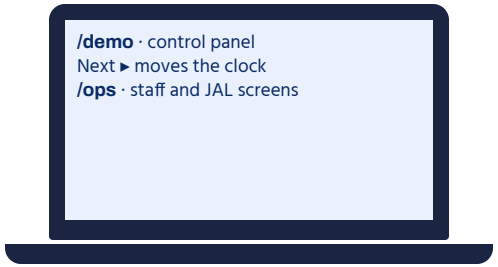
Measured on our build, simulated JL740 (186 passengers)

DEL NRT

GATE 07 LIVE DEMO

Four screens, one flight, about four minutes

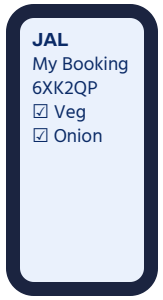
Everything runs live on our build. You can join the flight from your own phone.



PROJECTOR LAPTOP

The control room

Moves the story through time and shows the **staff, caterer and JAL** screens.



PRIYA'S PHONE

The passenger

Fills the meal form inside a **JAL-style booking page** and answers the fast-day question.



CREW IPAD · AIRPLANE MODE

The cabin crew

Downloads, goes **offline**, checks the galley and stops a wrong tray.



YOUR PHONE

You, the judges

Scan the QR, tell us your diet, and watch your seat light up on the crew iPad.

0 · 23 OCT 18:00
The 25-hour line

1 · 18:10
Priya's form

2 · 18:12
Fast-day question

3 · NIGHT
Chaos Mode: 7 problems

4 · 25 OCT 06:30
On board, offline

5 · LANDED
Results

Safety net: a "Do it for me" button runs any step if a click fails, and a recorded run is ready if the Wi-Fi drops.

Story smoke test passes all 8 steps on the current build

DEL ✈ NRT

GATE 08 WHY JAL WINS

What JAL gains, and how we keep JAL safe

WHAT JAL GAINS

Fewer wrong-meal incidents. Each one can cost ₹35,000–65,000 in court, plus bad press.

Proof in any dispute. A timestamped record of every step, so JAL can show what happened.

Happier Indian travellers, a market growing 35% a year, and JAL's "Customer Empathy" goal.

Less food waste. ₹152 a flight for sealed backups, instead of ₹2,359 of fresh meals thrown away.

Data no airline has: how often special meals fail, and where.

Nothing to rebuild. Same kitchens, same crew iPads, same booking page.

HOW WE KEEP JAL SAFE FROM LAWSUITS

Never edits JAL's booking system. It only reads, and a person approves every fix.

Never calls a meal "safe" for an allergy. It warns; crew and passenger read the label. JAL's own allergy meal (49 h) stays.

Rules decide, not AI. Every answer names its rule; anything unknown goes to a person.

Checked twice before serving: the seat screen asks, then the tray is scanned at the seat.

Every step is recorded with time and source, so blame is clear and fair.

Privacy by law: food rules, not religion; consent first; deleted after the trip.

"Today the airline listens once and hopes. ThaliBento keeps listening until the meal is served."

JAL pays a yearly licence per route.

PILOT: DELHI → TOKYO, READ-ONLY · MONTH 1 MEASURE · MONTH 2 ALERT · MONTH 3 ON BOARD · THEN: MORE ROUTES

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